MIL-R-81202D (AS)
3 February 1981
SUPERSEDING
MIL-R-81202C(AS)
3 February 1976

MILITARY SPECIFICATION

RECEIVER ASSEMBLY, NITROGEN,

FOR LAU-7 SERIES LAUNCHERS

This specification has been approved by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture, inspection and testing of nitrogen receiver assembly hereinafter called receiver assembly.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-N-411 Nitrogen, Technical.

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, New Jersey 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1055

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MILITARY

MIL-P-116	Preservation, Methods of.
DOD-D-1000	Drawing, Engineering and Associated Lists.
MIL-R-11468	Radiographic Inspection, Soundness Requirements for Arc and Gas Welds in Steel.
MIL-L-22620	Launcher, Guided Missile, LAU-7 Series.
MIL-I-23252	Indicator, Pressure.
MIL-V-23255	Valve, Pressure Relief.
MIL-V-23256	Valve, Seal.
MIL-V-85033	Valve, Charging.

STANDARDS

MILITARY

MIL-STD-109	Quality Assurance Terms and Definitions.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of.
MIL-STD-453	Inspection, Radiographic.
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of.
MIL-STD-810	Environmental Test. Methods.
MIL-STD-831	Test Reports, Preparation of.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND
(Code Ident 30003)

DL 58A164D556	Receiver Assembly, Nitrogen.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.4 and 6.4).

3.1.1 First article sample. Unless otherwise specified in the contract. or purchase order (see 6.2.1), a first article sample of five receiver assemblies shall-be delivered to the testing activity designated in the contract or purchase order (see 6.2.1). The first article sample shall be manufactured using

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the same methods, materials, processes and procedures proposed for production. Any production prior to acceptance of the first article sample is at the contractor's risk.

3.1.2 Disposition of preproduction samples. Preproduction samples procured under this specification are considered to be destructively tested and therefore are not to be included in net deliverable hardware.

3.2 Selection and quality verification of parts and materials. QualitY verified parts (see 3.2.1) and materials from approved sources (see 3.2.2) shall be used in the fabrication of the end item. The contractor shall prepare test reports of conformance to applicable requirements.

3.2.1 Approval of parts and materials. Parts and materials proposed for use in the fabrication of the end item that are not specified in detail herein or in applicable referenced documents shall be selected from other documents in the order of precedence specified in MIL-STD-143 and shall be approved by the procuring activity prior-to use. Approval shall be predicated upon source (vendor) approval and verification of part quality by test.

3.2.2 Source approval. The contractor shall be responsible for ascertaining that each proposed source for parts and materials maintains a quality control program to ensure consistent delivery of products conforming to applicable requirements. When parts or materials are fabricated by the contractor, the area used for that purpose shall be considered the same as an outside vendor insofar as source approval is concerned. Source approvals by the contractor shall be subject to concurrence by the procuring activity. Source approval will be granted by the procuring activity upon preparation of acceptable evidence of any of the following:

a. Formal on-site survey of the vendor's manufacturing facilities.

b. The source has been previously approved, and evidence is available to confirm that acceptable product quality has been maintained.

c. Other evidence of acceptable product quality is available.

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3.2.3 Verification of part quality. Quality verification shall be demonstrated by successful testing of the part to levels of environmental, electrical, and physical stresses equal to or greater than those which will be encountered in t-he system and as required by the part specification. Part quality may be demonstrated by any of the following:

a. When the part is manufactured and qualified to a military specification which includes requirements equal to or more severe than those of the end item application.

b. When the part is manufactured and qualified to a military specification not including all requirements of the application, and additional tests for such requirements have been performed.

3.2.4 Approved source/quality verified parts list (AS/QVPL) An AS/QVPL shall be prepared by the contractor for all subassemblies and parts for approval by the procuring activity. All entries on the AS/QVPL shall be fully substantiated by the preparation of data to satisfy the requirements of source approval and part test approval as specified herein. Reports of all quality verification testing shall conform to the requirements of MIL-STD-831. The contractor may solicit from the procuring activity information as to the status of quality verified parts and approved sources which may have been previously developed. Neither part environmental tests nor part environmental test reports are required in those instances where vendors are selected from current AS/QVPLs of other programs. Entries on the AS/QVPL shall identify the vendor and the product which is proposed for use in the receiver assembly by the contractor. In the event several sources have previously been approved for the application, the contractor shall specify the source of the parts to be used. Each item listed shall be supported in accordance with the provisions of this section. This listing does not supplant or modify the applicable documentation or other Government Qualified Parts Listing.

3.2.4.1 Conditions governing the quality verification of parts. Whenthe contractor proposes to demonstrate the quality verification of parts, as provided herein, the following provisions shall apply:

a. Test may be performed by the contractor, utilizing his own facilities, the facility of the manufacturer of the part or by any other suitable facility selected by the contractor. In any event, the facilities shall be approved by the contractor prior to the performance of any test. This information shall be prepared for approval by the procuring activity.

b. Prior to the performance of any test, a detailed test procedure shall be prepared by the contractor and approved by the procuring activity. The procedure shall identify, as a minimum, the step by step functions to be performed and the test equipment to be used, and shall provide for full conformance with the

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provisions of the applicable documentation. The Government reserves the right to review any of the individual test procedures at its discretion.

c. Quality verification testing may be monitored by a representative of the procuring activity.

3.2.4.2 Changes to the AS/QVPL. The contractor may develop alternative sources for any part, subject to the provisions herein. Proposals by the contractor for part substitution or changes to the AS/QVPL shall include proof that the proposed substitute part will meet the requirements of 3.2.2 and 3.2.3. The AS/QVPL shall be revised and updated as applicable to include approved changes.

3.2.4.3 Removal from the AS/QVPL. Evidence of poor quality may be cause for suspension or removal of a source from the AS/QVPL at the option of the contractor or the procuring activity (see 6.2.1). The contractor shall develop quality verified sources for those parts for which the approved source has been removed. Quality reverification shall be required under any of the following conditions unless otherwise directed by the procuring activity (see 6.2.1):

a. When the part has been modified by a change in design, materials, manufacturing process, manufacturing inspection procedures, or manufacturing facilities. The procuring activity shall be notified prior to delivery of any parts incorporating any such changes and be provided quantitative evidence of the effect of such changes on the performance or other characteristics of the receiver assembly.

b. When there is evidence that the quality of the part has not been maintained. This evidence may be in the form of accumulated failure reports which indicated that the quality of the part has deteriorated, or there is a failure trend.

c. When the manufacturer has not produced the item for 6 or more months.

3.2.5 Part conformance provisions. When a drawing or specification for a part invokes by reference the provisions of a military specification, total conformance to the provisions of the referenced specification shall be required. Total conformance is defined as conformance to the stated performance and product characteristics, and compliance with all quality assurance provisions, including qualification, quality verification, and quality conformance testing.

3.3 Design and construction. The design and construction of the receiver assembly shall be in accordance with the applicable drawings listed on DL 58A164D556 and as specified herein. The contractor shall notify the procuring

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activity of any changes in the design and construction after the receiver assembly has been accepted by the procuring activity. The receiver assembly shall consist of the following components:

3.3.1 Nitrogen receiver. The nitrogen receiver shall be in accordance with the requirements stipulated herein.

3.3.2 Gas pressure indicator. The gas pressure indicator shall be in accordance with MIL-I-23252.

3.3.3 Charging valve. The charging valve shall be in accordance with MIL-V-85033.

3.3.4 Gas pressure relief valve. The gas pressure relief valve shall be in accordance with MIL-V-23255.

3.3.5 Seal valve. The seal valve shall be in accordance with MIL-V-23256.

3.3.6 Filter cartridge assembly. The filter cartridge assembly shall be in accordance with the applicable requirements of MIL-V-23256.

3.3.7 Materials. Materials shall be selected on the basis of strength, workability and compatibility with other materials in the item so as to minimize galvanic action, corrosion, etc.

3.4 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Changes in part numbers shall be governed by drawing number requirements of DOD-D-1000.

3.5 Identification of parts. All parts shall be marked as specified on DL 58A164D556, or in accordance with MIL-STD-130 if not specified on the drawing.

3.6 Nitrogen supply gas. The gas used in the receiver assembly shall be in accordance with BB-N-411, Type 1, Class 1, Grade A, except that the maximum contaminant level shall not exceed the levels shown in Table I.

3.7 Performance. The receiver assembly shall, when assembled in accordance with the drawings listed on DL 58A164D556, meet the requirements as specified herein.

3.7.1 Radiological. The nitrogen receiver shall meet the radiological requirements of MIL-R-11468 interpreted in accordance with MIL-STD-453.

3.7.2 Permanent volumetric expansion. The percentage of permanent volumetric expansion shall not exceed five percent of the total volumetric expansion, after the nitrogen receiver has been pressurized hydrostatically to 5500 ±100 pounds per square inch gage (psig).

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TABLE I. Nitrogen supplv gas - acceptable contaminant level

Contaminant	Maximum Contaminant Level
Water (H_20)	2 parts per million (by volume)
Carbon Dioxide	2 parts per million (by volume)
Freon	3 parts per million (by volume)
Total hydrocarbons (including Methane and Acetylene)	10 parts per million (by volume)
Acetylene	1 part per million (by volume)
Methane	6 parts per million (by volume)
Hydrocarbons (others)	3 parts per million (by volume)
Foreign particles	Maximum of 100 particles, 5.0 to 10.0 microns per cubic foot of flow. No particle larger than 10.0 microns allowed.

3.7.3 Visual. The nitrogen receiver shall meet the visual requirements to ensure conformance to all requirements listed on DL 58A164D556 and as specified herein (see 4.6.3).

3.7.4 Cleanliness. The nitrogen receiver shall be pressurized through the charging valve to 2000 ±50 psig with nitrogen gas (see 3.6). The samples taken from the nitrogen receiver shall not exceed the maximum contaminant levels -specified in Table I, after exposure to the temperature conditioning cycle as follows:

a. The ambient temperature shall be reduced to 15.6°C (60°F) and maintained for approximately 12 hours.

b. The ambient temperature shall be increased to 43.3°C (110°F) and maintained for approximately 12 hours.

c. The ambient temperature conditioning of a and b shall be repeated.

At the completion of the conditioning cycle, and within 10 minutes after removal from the 43.3°C (110°F) ambient condition, a gas sample shall be withdrawn from the receiver assembly charging valve.

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3.7.5 Functional. The receiver assembly shall operate satisfactorily after meeting the following requirements:

3.7.5.1 Proof pressure. There shall be no failure of parts to the receiver assembly after being pressurized to 5500 +/-100 psig for three minutes. The pressure relief valve shall not be utilized for this inspection. A suitable pressure fitting shall be used instead of the pressure relief valve.

3.7.5.2 Leakage. There shall be no leakage of the receiver assembly in excess of 0.5 cubic centimeters per minute, after three minutes exposure to each of the following pressures:

a. 200 psig

b. 1000 psig

c. 2200 psig

d. 3000 psig

e. 3800 psig

3.7.5.3 Operating pressure. The receiver shall meet the dimensional requirements for the external diameter of its transverse section when subjected to an internal pressure of 3800 -0 psig.

3.7.6 Environmental. The receiver assembly shall operate satisfactorily after exposure to the following requirements:

3.7.6.1 High temperature. The receiver assembly shall operate satisfactorily and meet the requirements of 3.7.5.1 and 3.7.5.2 after the high temperature test of MIL-STD-810, Method 501, Procedure I, except the temperature shall be 121° Celsius (C) [250° Fahrenheit (F)].

3.7.6.2 Low temperature. The receiver assembly shall operate satisfactorily and meet the requirements of 3.7.5.1 and 3.7.5.2 after the low temperature test. of MIL-STD-810, Method 502, Procedure I, for a period of four hours. The storage temperature shall be -54°C {-65°F).

3.7.6.3 Thermal shock. The receiver assembly shall operate satisfactorily and meet the requirements of 3.7.5.1 and 3.7.5.2 after the test of MIL-STD-810, Method 503, Procedure I.

3.7.6.4 Salt fog. The receiver assembly shall operate satisfactorily and meet the requirements of 3.7.5.1 and 3.7.5.2 after the test of MIL-STD-810, Method 509, Procedure I, except the duration shall be 50 hours.

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3.7.6.5 Humidity. The receiver assembly shall operate satisfactorily and meet the requirements of 3.7.5.1 and 3.7.5.2 after the test of MIL-STD-810, Method 507, Procedure I, except that four 24 hour cycles shall be substituted for the ten 24 hour cycles.

3.7.6.6 Vibration. The receiver assembly shall operate satisfactorily and meet the requirements of 3.7.5.1 and 3.7.5.2 after the test of MIL-L-22620. The receiver assembly shall be mounted to a LAU-7 series launcher for this inspection.

3.7.6.7 Impact. The receiver assembly shall not sustain critical physical damage or have a loss of pressure after being subjected to three free-fall drops from a height of five feet.

3.7.7 Burst pressure. There shall be no failure of parts of the receiver assembly after being pressurized to 6700 +/-100 psig for three minutes. The pressure relief valve shall be removed and a suitable pressurization fitting used in its place. Deterioration of the pressure indicator accuracy shall not constitute a failure.

3.7.8 Service life. The receiver assembly shall operate satisfactorily and meet the requirements of 3.7.5.1 and 3.7.5.2 after the service life tests described as follows. The pressure relief valve shall be removed and a suitable pressurization fitting used in its place.

a. A total of 500 cycles, as follows, at a maximum rate of 6 cycles per minute:

(1.) Pressurize receiver to 225 +/-50 psig.

(2.) Increase pressure uniformly to 4000 +/-250 psig.

(3.) Exhaust uniformly to 225 +/-50 psig.

b. A total of 4000 cycles, as follows, at a maximum rate of 9 cycles per minute:

(1.) Pressurize receiver to 2500 +/-150 psig.

(2.) Increase pressure uniformly to 3500 +/-200 psig.

(3.) Exhaust uniformly to 2500 +/-150 psig.

c. A total of 500 cycles, as follows, at a maximum rate of 6 cycles per minute:

(1.) Pressurize receiver to 225 +/-50 psig.

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(2.) Increase pressure uniformly to 4000 +/-250 psig.

(3.) Exhaust uniformly to 225 +/-50 psig.

3.8 Precedence. The selection of specifications and standards for use in the design and development shall be in accordance with the requirements of MIL-STD-143. When the requirements, this specification, and referenced documents are in conflict, the following precedence shall apply:

a. This specification. This specification shall have precedence over all referenced documents.

b. Referenced documents. Any documents referenced in this specification shall have precedence over all subsidiary documents referenced therein.

3.9 Workmanship. The workmanship displayed in fabrication and assembly of the receiver assembly shall be such as to assure, within design limitations, ability of the receiver assembly to meet their performance requirements under all applicable environmental conditions specified herein. Particular attention shall be paid to neatness and thoroughness of assembly, marking of parts and assemblies, plating, painting, machining, welding, brazing and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in the approved first article sample, subject to any qualification stated in the government's notice of approval, shall determine the requirements of the remainder of the contract relative to workmanship.

4. QUALITY ASSURANCEE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure that the supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection of the receiver assemblies shall be classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

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4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed under the following conditions and tolerances:

a. Room temperature. Room ambient of 25 +/-10°C (77 +/-18°F) indicated.

b. Test temperature. All parts of the test item shall be stabilized at the specified temperature +/-2.8°C (+/-5°F) indicated prior to conducting any operational tests. Unless otherwise specified herein, temperature stabilization will have been attained when the indicated temperature of the surface of the largest mass of the test item does not change more than +/-2.8°C (+/-5°F) in a period of one hour.

c. Humidity. Room ambient to 95 percent relative humidity.

d. Altitude. Unless otherwise specified herein, testing shall be performed at normal ground elevation.

e. Test gas. When conducting the testing of 4.6.4, the test gas shall be nitrogen in accordance with 3.6. For all other tests, either nitrogen conforming to the requirements of BB-N-411 Type I, Class 1, Grade A or air of equivalent purity may be used.

f. Test attitude. Unless otherwise specified herein, all tests shall be conducted with the longitudinal axis of the receiver assembly in a horizontal plane. Hydrostatic testing may be conducted in any attitude.

g. Torque requirements. The installation torque for mating the receiver assembly to the control valve shall be hand-tight. The installation torques for mating the subassemblies to the *nitrogen* receiver shall be as listed in 6.5.

h. Volume and flow. Where required, all volume and flow measurements shall be corrected to normal temperature and barometric pressure (NPT) conditions. NPT conditions are 760 millimeters (29.92 inches) of mercury and 21.1°C (70°F).

4.4 First article inspection. First article inspection shall consist of the tests outlined in Table 11. The procuring activity reserves the right to delete tests and alter the test sequence of Table II (see 6.2.1). First article approval is valid only on the contract or purchase order under which it is granted, unless extended by the government to other contracts or purchase orders.

4.4.1 Receiver assembly samples. First article receiver assembly samples shall consist of the first five receiver assemblies manufactured on the contract or purchase order. The receiver assemblies shall be permanently labeled

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TABLE II. First article inspections - schedule and sequence 1/.

Test	Title	Test requirement paragraph	Test method paragraph	Sample				
				A	B	C	D	E
A	Radiographic inspection 2/	3.7.1	4.6.1	1	1	1	1	1
B	Permanent volumetric expansion 3/	3.7.2	4.6.2	2	2	2	2	2
C	Inspection	3.7.3	4.6.3	3	3	3	3	3
D	Cleanliness	3.7.4	4.6.4	4	4	4	4	4
E	Proof pressure	3.7.5.1	4.6.5.1	5	5	5	5	5
F	Leakage	3.7.5.2	4.6.5.2	6	6	6	6	6
G	Operating pressure	3.7.5.3	4.6.5.3	7	7	7	7	7
H	High temperature	3.7.6.1	4.6.6.1	8	x	x	x	x
I	Low temperature	3.7.6.2	4.6.6.2	9	x	9	x	x
J	Thermal shock	3.7.6.3	4.6.6.3	x	8	x	8	x
K	Salt fog	3.7.6.4	4.6.6.4	x	9	x	9	x
L	Humidity	3.7.6.5	4.6.6.5	x	10	x	10	x
M	Vibration	3.7.6.6	4.6.6.6	10	x	10	x	x
N	Impact	3.7.6.7	4.6.6.7	x	x	x	11	x
O	Burst pressure	3.7.7	4.6.7	x	x	11	x	x
P	Service life 4/	3.7.8	4.6.8	x	11	x	x	x

1/ Test sequence is denoted by the numbers in the sample columns. An X denotes test not required.

2/ Radiographic inspection shall be performed by the contractor prior to filament winding. This test will not be conducted by the government approved test laboratory unless directed by the procuring activity (see 6.2.1).

3/ Permanent volumetric expansion shall be performed by the contractor prior to mounting components. This test will not be conducted by the government approved test laboratory unless directed by the procuring activity (see 6.2.1).

4/ Receiver assemblies subjected to life tests shall not be subjected to the burst test.

for ease of identification and shall be produced with the equipment and procedures that will be used for the production of all receiver assemblies designated in the contract or purchase order. Prior to submission, the contractor

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shall inspect the samples to the degree necessary to assure that they conform to the requirements of the contract and prepare a first article test report of this inspection with the sample, including statements of findings and certificates of conformance for materials and tests A through G of Table II. Replacement relief valve o-rings shall be supplied for receiver assembly samples experiencing proof pressure testing.

4.5 Quality conformance inspection. Quality conformance inspection consists of those tests conducted on receiver assemblies submitted for acceptance under the contract or purchase order. Acceptance of receiver assemblies shall be on an inspection lot basis. Quality conformance inspection shall consist of the following tests:

a. Individual tests (see Table III, Group 1).

b. Sampling tests (see Table III, Groups 1 and 2).

4.5.1 Individual tests. Individual tests are those tests conducted by the contractor on each receiver assembly submitted for acceptance under the contract or purchase order. Individual tests for. receiver assemblies are shown in Table III, Group 1. Failure to pass these tests shall be cause for rejection.

4.5.2 Sampling tests. A random sampling of receiver assemblies shall be selected by the procuring activity from each inspection lot in accordance with Table IV and shall be subjected to the tests listed in Table 111, Groups 1 and 2. Group 1 tests shall be conducted prior to Group 2 tests. Failure to pass these tests shall be cause for rejection of the lot. The procuring activity reserves the right to delete tests and alter the test sequence of Table III (see 6.2.1). In the event any sample representing an inspection lot fails to meet the inspection and test requirements outlined herein, the contractor, at the discretion of the procuring activity, may have the opportunity of reworking the entire affected lot (see 6.2.1). The procuring activity shall determine which inspections and tests of this specification are to be performed on the reworked inspection lot samples (see 6.2.1). The number of test samples shall be at the discretion of the procuring activity (see 6.2.1). Replacement relief valve o-rings shall be supplied for receiver assembly samples experiencing proof pressure testing.

4.6 Inspection methods.

4.6.1 Radiographic inspection. All the welds of the nitrogen receiver shall be subjected to a radiographic inspection in accordance with MIL-R-11468 and interpreted in accordance with MIL-STD-453. Radiographic inspection shall be performed by the contractor prior to filament winding. This test will not be conducted by the government approved test laboratory unless directed by the procuring activity.

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TABLE III. Quality conformance inspections - schedule and sequence[1/].

Test	Test requirement paragraph	Test method paragraph	Sample			
GROUP 1			ALL SAMPLES			
Radiographic inspection[2/]	3.7.1	4.6.1	1			
Permanent volumetric expansion[3/]	3.7.2	4.6.2	2			
Inspection	3.7.3	4.6.3	3			
Cleanliness	3.7.4	4.6.4	4			
Proof pressure	3.7.5.1	4.6.5.1	5			
Leakage	3.7.5.2	4.6.5.2	6			
Operating pressure	3.7.5.3	4.6.5.3	7			
GROUP 2[4/]			A	B	C	D
High temperature	3.7.6.1	4.6.6.1	1	x	1	x
Low temperature	3.7.6.2	4.6.6.2	2	x	2	x
Thermal shock	3.7.6.3	4.6.6.3	x	1	x	1
Salt fog	3.7.6.4	4.6.6.4	x	2	x	2
Humidity	3.7.6.5	4.6.6.5	x	3	x	3
Vibration	3.7.6.6	4.6.6.6	3	x	3	x
Impact	3.7.6.7	4.6.6.7	x	x	x	4
Burst pressure	3.7.7	4.6.7	x	x	4	x
Service life[5/]	3.7.8	4.6.8	x	4	x	x

1/ Test sequence is denoted by the number in the sample columns. An X denotes tests not required.

2/ Radiographic inspection shall be performed by the contractor prior to filament winding. This test will not be conducted by the government approved test laboratory unless directed by the procuring activity (see 6.2.1).

3/ Permanent volumetric expansion shall be performed by the contractor prior to mounting components. This test will not be performed by the government approved test laboratory unless directed by the procuring activity (see 6.2.1).

4/ The procuring activity shall decide which tests of Table III, Group 2, each additional receiver assembly in excess of four shall receive (see 6.2.1).

5/ Receiver assemblies subjected to life tests shall not be subjected to the burst test.

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TABLE IV. Sampling test items.

Lot size	Number of samples to be tested
499 or less	4
500 or more	4 plus 1 for each additional 500 or fraction thereof.

4.6.2 Permanent volumetric expansion. The nitrogen receiver shall be pressurized hydrostatically to 5500 +/-100 psig and the total volumetric expansion shall be determined. The pressure shall then be reduced to atmospheric and the permanent volumetric expansion determined. The percentage of permanent volumetric expansion shall not exceed 5 percent of the total volumetric expansion. Test values for all parameters shall be prepared for review by the procuring activity. The nitrogen receiver shall be tested without its subassemblies installed. All open ports shall be sealed and a pressurization fitting installed. Testing may be conducted in any attitude. This test will not be performed by the Government-approved test laboratory unless directed by the procuring activity (see 6.2.1).

4.6.3 Inspection-receiver assembly. The receiver assembly shall be inspected as follows:

4.6.3.1 Individual tests-visual. The receiver assembly shall be inspected to ensure that it has been properly assembled and adjusted, and that the workmanship is as specified herein.

4.6.3.2 First article and quality conformance tests.

a. The receiver assembly shall be thoroughly examined to determine conformance to all the requirements of the drawings listed on DL 58A164D556 and this specification. Prior to performing the cleanliness test, only external dimensioning shall be performed.

b. The receiver assembly shall be examined to ascertain that the packaging, packing and marking is in accordance with Section 5.

4.6.4 Cleanliness. The receiver assembly shall be pressurized through the charging valve to 2000 +/-50 psig with nitrogen gas (see 3.6). Initial back ground contamination levels of the test gas shall be recorded. These levels

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shall constitute tare values, used to correct the final contamination levels. The receiver assembly shall then be subjected to a temperature conditioning cycle as follows:

a. The ambient temperature shall be reduced to 15.6°C (60°F) and maintained for approximately 12 hours.

b. The ambient temperature shall be increased to 43.3°C (110°F) and maintained for approximately 12 hours.

c. The ambient temperature conditioning of a and b shall be repeated.

At the completion of the conditioning cycle, and within 10 minutes after removal from the 43.3°C (110°F) ambient condition, a gas sample shall be withdrawn from the receiver assembly charging valve. This sample shall be used to determine water contamination. For consistency, the water sample must be taken at the 2000 psig level. Subsequent samples would be taken at random lower pressures to determine remaining contaminant levels of Table I. The levels shall not exceed the requirements of that table.

4.6.5 Functional test. The receiver assembly shall be subjected to the following tests.

CAUTION

Do not pressurize receiver assemblies at high' fill rates. Filler valves and receiver valves fitted with integral or poppet-type valve seats may become damaged due to overheating. Do not lubricate rubber or plastic seals (O-rings, valve seats, etc.), threads or other mating parts that might, thereby, contaminate the gas system.

4.6.5.1 Proof pressure. The receiver assembly shall be pressurized to 5500 +/-100 psig for three minutes. To conduct this test the pressure relief valve shall be removed from the nitrogen receiver and a suitable pressure fitting installed. This fitting shall be utilized for the pressurization. There shall be no failure of parts. In addition, when the receiver assembly is returned to atmospheric pressure there shall be no visual indication of permanent deformation. Replace the original relief valve o-ring with a new contractor supplied o-ring, before commencing with the next test.

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4.6.5.2 Leakage. The receiver assembly shall be checked for leakage at the following pressures for three minutes:

a. 200 psig

b. 1000 psig

c. 2200 psig

d. 3000 psig

e. 3800 psig

Leakage shall not exceed 0.5 cubic centimeters per minute at any of the pressures between 200 and 3800 psig.

4.6.5.3 Operating pressure. The nitrogen receiver shall meet the dimensional requirements for the external diameter of its transverse section at atmospheric pressure and at an internal pressure of 3800 +100/-0 psig.

4.6.6 Environmental tests. The receiver assemblies shall be subjected to the following environmental tests.

4.6.6.1 High temperature. The receiver assembly shall be subjected to the high temperature test in accordance with MIL-STD-810, Method 501, Procedure I, except that the temperature shall be 121°C (250°F). While subjected to this temperature, the pressure in the receiver assembly shall be maintained at 3800 +0 -200 psig. At the completion of the test, the receiver assembly shall be visually inspected for damage and discoloration, and tested in accordance with 4.6.5.2 and 4.6.5.3.

4.6.6.2 Low temperature. The receiver assembly shall be subjected to the low temperature test in accordance with MIL-STD-810, Method 502, Procedure I, for a period of four hours. The storage temperature shall be -54°C (-65°F). The receiver assembly shall initially be pressurized to 3800 +/-100 psig and sealed. After completion of the exposure, the receiver assembly shall be allowed to stabilize at room temperature. Then the internal pressure shall be accurately measured. The pressure loss shall not exceed 300 psig. The receiver assembly shall be visually inspected for damage and discoloration, and tested in accordance with 4.6.5.2 and 4.6.5.3.

4.6.6.3 Thermal shock. The receiver assembly shall be subjected to the temperature shock test in accordance with MIL-STD-810, Method 503, Procedure I. Prior to being subjected to the temperature conditions, the receiver assembly shall be pressurized to 3000 +/-100 psig. At the completion of the test, the receiver assembly shall be visually inspected for damage and discoloration, and tested in accordance with 4.6.5.2 and 4.6.5.3.

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4.6.6.4 Salt fog. The receiver assembly shall be subjected to the salt fog test in accordance with MIL-STD-810, Method 509, Procedure I except that the duration shall be 50 hours. All fittings normally connected shall be protected during climatic conditioning. The receiver shall be pressurized to 225 +/-25 psig. At the completion of the test, the receiver assembly shall be visually inspected for damage, corrosion and discoloration, and tested in accordance with 4.6.5.2 and 4.6.5.3.

4.6.6.5 Humidity. The receiver assembly shall be subjected to the humidity test in accordance with MIL-STD-810, Method 507, Procedure I, except that four 24-hour cycles shall be substituted for the ten 24-hour cycles. All fittings normally connected shall be protected during climatic conditioning. The receiver assembly shall be pressurized to 225 +/-25 psig. At the completion of the test, the receiver assembly shall be visually inspected for damage, corrosion and discoloration, and tested in accordance with 4.6.5.2 and 4.6.5.3.

4.6.6.6 Vibration. The receiver assembly shall be subjected to the vibration test in accordance with MIL-L-22620 while mounted in a LAU-7 series launcher. The receiver assembly shall be pressurized to 3000 +/-100 psig. After testing is completed for each axis of vibration, the receiver assembly shall be visually inspected for damage and tested in accordance with 4.6.5.2 and 4.6.5.3.

4.6.6.7 Impact. The receiver assembly shall be subjected to three free-fall drops from a height of five feet while pressurized to 3800 +/-100 psig. The free-fall height shall be measured from the geometric center of the receiver assembly to the floor, which shall be a rigid concrete surface. Prior to each impact, an 8-inch I-beam at least six feet long shall be placed upright on the floor (so that one flat side of the I-beam lies on the floor). The geometric centers of the I-beam and receiver assembly shall be located in-line vertically, but their longitudinal axes shall be positioned in normal planes. Extreme care shall be taken during this test. The unit shall be released using a remote actuated device and provisions shall be made to allow remote venting of the pressure immediately after each release. The attitude condition for each drop, as referenced to the longitudinal axis (see Figure 1), shall be as follows:

a. First impact - aft end down 45° to horizontal.

b. Second impact - forward end down 45° to horizontal.

c. Third impact - horizontal.

After each impact, the receiver assembly shall be rotated 120° circumferentially (see Figure I). There shall be no loss of pressure during this test and the receiver assembly shall not sustain critical physical damage. This test is considered a destructive test (see 3.1.2).

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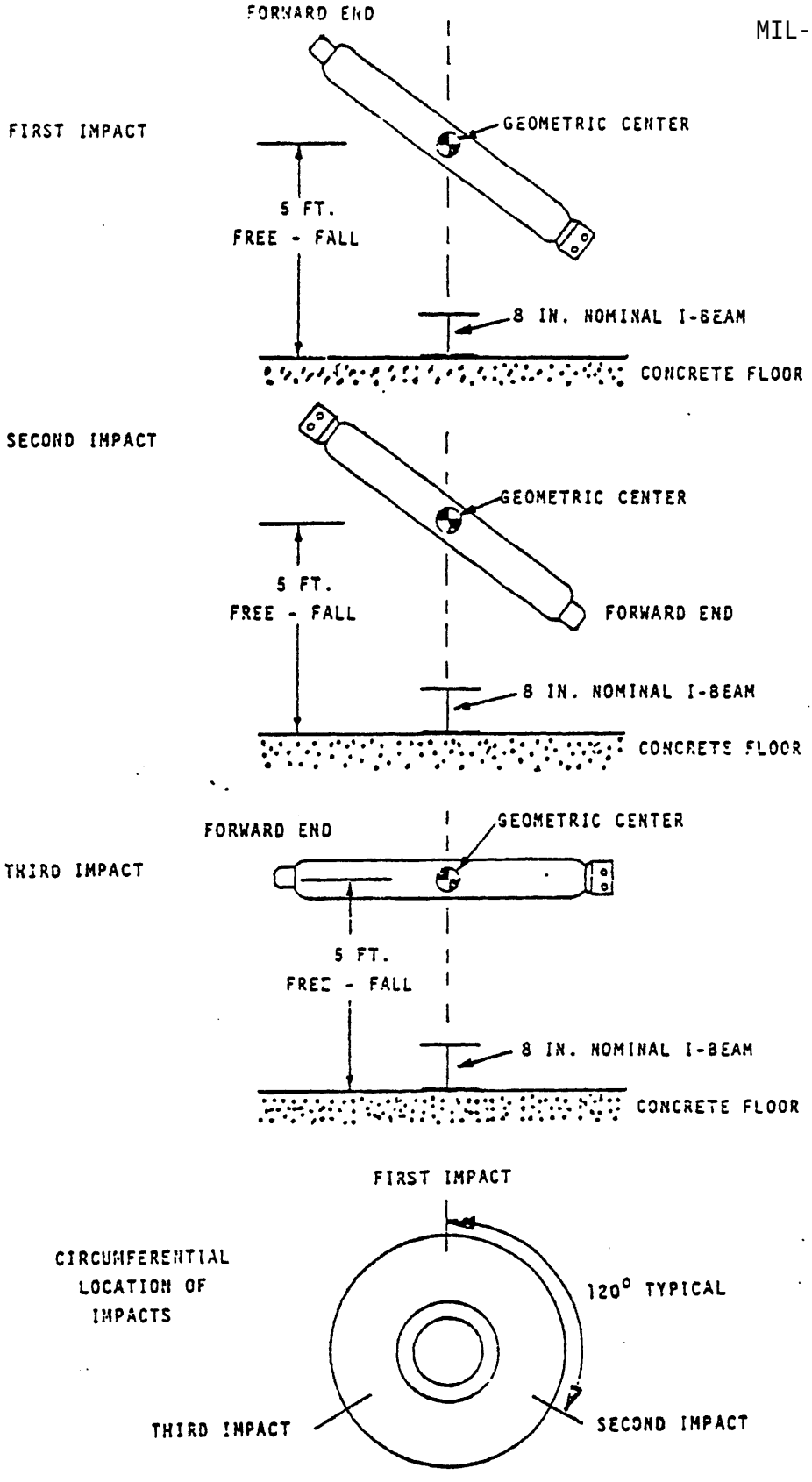


FIGURE 1. Impact test-receiver assembly attitude.

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4.6.7 Burst pressure. The receiver assembly shall be hydrostatically pressurized to 6700 +/-100 psig for three minutes. To conduct this test, the pressure relief valve shall be removed from the nitrogen receiver and a suitable pressurization fitting installed. There shall be no failure of parts. Deterioration of the pressure indicator accuracy shall not constitute a failure. This is considered a destructive test (see 3.1.2).

4.6.8 Service life. The receiver assembly shall be subjected to the following service life test performed hydrostatically:

a. A total of 500 cycles, as follows, at a maximum rate of 6 cycles per minute:

(l.) Pressurize receiver to 225 +/-50 psig.

(2.) Increase pressure uniformly to 4000 +/-250 psig.

(3.) Exhaust uniformly to 225 +/-50 psig.

b. A total of 4000 cycles, as follows, at a maximum rate of 9 cycles per minute:

(l.) Pressurize receiver to 2500 +/-150 psig.

(2.) Increase pressure uniformly to 3500 +/-200 psig.

(3.) Exhaust uniformly to 2500 +/-150 psig.

c. A total of 500 cycles, as follows, at a maximum rate of 6 cycles per minute:

(1.) Pressurize receiver to 225 +/-50 psig.

(2.) Increase pressure uniformly to 4000 +/-250 psig.

(3.) Exhaust uniformly to 225 +/-50 psig.

To conduct this test, the pressure relief valve shall be removed from the nitrogen receiver and a suitable pressurization fitting installed. At the completion of the test, the nitrogen receiver shall be visually inspected for damage and tested in accordance with 4.6.5.2 and 4.6.5.3.

5. PREPARATION FOR DELIVERY

5.1 Receiver assembly preservation and packaging. Preservation and packaging shall be in accordance with Level A or C of MIL-STD-794, as specified in the contract or purchase order (see 6.2.1).

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5.1.1 Level A. The receiver assemblies shall be unit packaged, one each, in accordance with Method II of MIL-P-116, without the use of the preservation compound. Immediately after installation of the filter cartridge, each receiver assembly shall be capped with an external seal cap, Drawing 58A164B806. Receiver assemblies shall be pressurized at 225 +/-25 psig with nitrogen conforming to 3.6.

5.1.2 Level C. Receiver assemblies shall be unit packaged, one each, in accordance with Method II of MIL-P-116, without the use of preservation compound . Immediately after installation of the filter cartridge, each receiver assembly shall be capped with an external seal cap. Receiver assemblies shall be pressurized at 225 +/-25 psig with nitrogen conforming to 3.6.

5.2 Packing. Shipping containers shall be suitable for Level A or Level C packing as specified in MIL-STD-794.

5.3 Unitized loading. Unitized loading for the receiver assemblies is authorized as specified in MIL-STD-794.

5.4 Marking. All shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The nitrogen receiver assembly is intended for use in storing and delivering, upon demand, high purity nitrogen gas. The receiver assembly is capable of operating within the temperature range of -54°C to +121°C (-65°F to +250°F).

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, revision letter and date of this specification.

b. Government-approved test laboratory (see 3.1.1).

c. First article requirement (see 3.1 and 3.1.1).

d. Option for removal from AS/QVPL (see 3.2.4.3).

e. Option for quality reverification (see 3.2.4.3).

f. Responsibility for inspection (see 4.1).

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g. Samples subjected to sampling inspection (see 4.5.2) shall not be considered or accepted as part of the contract or purchase order (see 3.1.2 and 4.6.7).

h. Rework and restest requirements and quantity to be submitted for inspection lot samples (see 4.5.2).

i. Tests deleted and altered test sequence (see 4.4, 4.5.2, footnote 4/ of Table III).

j. Tests to be performed by Government-approved test laboratory if directed by procuring activity (see 4.6.1, 4.6.2, footnotes 2/ and 3/ of Tables II and III).

k. Selection of applicable levels of preservation, packaging and packing required (see 5).

l. Government-approved test laboratory (see 3.1.1).

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
3.2, 4.6.2	Test Reports	DI-T-21466A
3.2.1, 3.2.2, 3.2.4 & 3.2.4.2	Approved Parts Source	UDI-R-20331B
3.2.4.1	Test Procedures	DI-T-3702A
3.3	Engineering Change Proposal	DI-E-2037
4.4.1	Report, First Article Test	UDI-T-23790

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Quality assurance definitions. Definitions for quality assurance terms are in accordance with MIL-STD-109.

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The first article should be a first production item. The

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first article should consist of five receiver assemblies. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test and approval of the first article.

6.5 Contract or purchase order provision. Contracts or purchase orders should specify the following provision for first article inspection. "When a contractor is in continuous production of the receiver assemblies applicable to this specification from contract to contract, consideration shall be given to waive the first article inspections. However, waiver of first article inspections shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract." The words "purchase order" should be substituted throughout for the word "contract" when applicable.

6.6 Subassembly torque requirements. The torque values for the installation of the subassemblies to the nitrogen receiver are as follows:

a.	Relief valve	225 +/-25 inch pounds
b.	Pressure indicator	100 +/-25 inch pounds
c.	Filter cartridge fitting	250 $^{+25}_{-0}$ inch pounds
d.	Seal valve	450 +/-25 inch pounds
e.	Charging valve	275 +/-25 inch pounds

The charging valve closing torque is 50 +/-5 inch pounds.

6.7 Cleaning and assembly procedure. The following procedure is suggested for cleaning and assembling LAU-7 nitrogen receivers. The procedure entails a filtered liquid flush to remove gross particulate matter, followed by a vacuum-heat-pressure gas purge to remove final particulate matter and gaseous contamination. The parts are cleaned of particulate material prior to installation and are cleaned of gaseous contamination with the receiver. The procedure is most effective if carried out in one continuous operation through 6.6.2.2.6. If this is not done, it is necessary to store flushed receivers and parts in such a reamer as to prevent particle contamination.

6.7.1 Precleaning.

a. If manufactured and packaged in accordance with cognizant military specification, parts are precleaned as supplied.

b. The receiver shall be precleaned to remove machining oils, residues and scale using standard commercial procedures that will not damage the interior lining material.

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6.7.2 Liquid flush.

6.7.2.1 Parts.

6.7.2.1.1 Type of liquid. It is suggested that parts be flushed with freon TF or 113, isopropyl alcohol or ethyl alcohol.

6.7.2.1.2 Type of filter. The manufacturer may choose the type of filter used, but it is suggested that the membrane filter be rated at 1.2 microns or smaller, and that the material be compatible with the liquid used.

6.7.2.1.3 Flushing. Filtered liquid (see 6.6.2.1.1 and 6.6.2.1.2) shall be used to rinse the portions of the parts that extend into the receiver. The charging valve shall be opened and rinsed.

6.7.2.1.4 Prepared parts. Prepared parts shall be immediately assembled into the receiver. If this is not practicable, parts shall be stored in such a manner as to prevent recontamination. Specially prepared plastic bags, laminar flow benches or special enclosures supplied with filtered purge air may be used for this purpose.

6.7.2.2 Receiver.

6.7.2.2.1 Type of liquid. Water or freon TF or 113 may be used to flush the receiver. For rinsing purposes, either isopropyl alcohol, ethyl alcohol or petroleum ether may be used. .

6.7.2.2.2 Type of filter. The manufacturer may choose the type of filter used, but it is suggested that the membrane filter be rated at 1.2 microns or smaller, and that the material be compatible with the liquid used.

6.7.2.2.3 Inspection procedure. The manufacturer may choose the inspection procedures employed, but it is suggested that the procedures include a visual microscopic examination of the gridded membrane filter, after a measured sample of flushing liquid from the receiver has been passed through the filter. Published standard counting procedures should be used in making microscopic examinations (i.e., ARP598A published by The Society of Automotive Engineers).

6.7.2.2.4 Liquid flush acceptance criteria. Because of set-up variations, acceptance criteria shall be determined experimentally by the manufacturer. It is not necessary nor expected that the final specification requirements be met at this point in the cleaning procedure: it is only necessary to establish that the receiver is sufficiently free of particles to allow the final steps of the procedure to complete the cleaning process.

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6.7.2.2.5 Flushing procedure.

a. The liquid (6.6.2.2.1) shall be circulated through the filter, into the receiver, out of the receiver and either down the drain or through the pump back to the filter.

b. Flushing shall be continued until the sample taken using the suggested inspection procedure (6.6.2.2.3) meets the acceptance criteria.

c. The liquid shall be emptied from the receiver.

d. The receiver shall be rinsed with filtered solution (see 6.6.2.2.1 and 6.6.2.2.2).

6.7.2.2.6 Prepared parts. Prepared parts shall be immediately assembled into the receiver. The tab washer shall be bent and the safety-wire installed.

6.7.3 Heat-vacuum-gas purge cycle. The receiver assembly shall be subjected to a heat-vacuum-gas purge cycling test. The temperature of the receiver during the cycle shall be 110 +/-5.5°C (230 +/-10°F). The vacuum in the receiver during the cycle shall be 25 +/-5 microns of mercury, maximum pressure. The gas pressure shall be 1500 +/-50 psig minimum, and the gas shall meet the same requirements as established for the nitrogen gas used to test the receivers. Air may be used in lieu of nitrogen. The test shall be conducted as follows:

a. The receivers shall be heated to test temperature.

b. After temperature stabilization, receivers shall be subjected to three cycles of:

(1.) Ten minutes vacuum.

(2.) Five minutes pressure

c. At the end of the third cycle, exhaust the receiver to 225 psig and immediately pressurize to 2500 psi g using nitrogen.

d. Exhaust the receiver to 500 psig and allow to cool.

The receiver shall be monitored for cleanliness in accordance with the manufacturer's standard procedure.

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6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to previous issue, due to the extensiveness of the changes.

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